

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)**

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 13	Team No- 10	Project Title - Real estate listing & virtual tour portal
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1. Abstract

Estate – Real Estate Listing & Virtual Tour Portal is a full-stack web application designed to simplify and modernize the process of discovering, evaluating, and enquiring about real estate properties. The platform initially targets the Hyderabad real estate market and connects property buyers with real estate agents through a centralized digital platform.

Estate enables users to search and explore properties using location-based map search and advanced filters such as price, property type, BHK configuration, area, and amenities. Each property can include detailed information, images, location details, and an immersive **360° virtual tour**, allowing buyers to experience a property remotely before arranging a physical visit. The platform also provides an **EMI calculator** to help users estimate monthly loan payments and make more informed purchasing decisions.

For real estate agents, Estate provides an **Agent Dashboard** to create and manage property listings, monitor enquiries, and manage potential customer leads. An **Admin module** supports property and user management and helps maintain listing quality through a verification process.

The application is developed using **React.js** for the frontend, **Node.js and Express.js** for backend services, and **MySQL** for structured data management. **Map box** is integrated to provide interactive map-based property discovery, while a 360° viewer enables virtual property exploration. RESTful APIs connect the frontend with backend services, providing a scalable architecture for future development.

The long-term vision of Estate is to evolve into a smart real estate platform by incorporating AI-based property recommendations, price prediction, locality insights, and personalized property discovery. By combining verified listings, interactive maps, virtual tours, financial tools, and agent lead management, estate aims to make property discovery more transparent, efficient, and convenient for buyers while providing valuable digital tools for real estate agents.

2. Team Details

Roll Number(s)	Name of the Student	Signature
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3. Frameworks / Platforms / Software Used

The EstateX platform uses a combination of modern web development frameworks, database technologies, APIs, and development tools to provide a scalable and interactive real estate experience.

1. Technology / Software	Purpose
React.js	Used to develop the interactive and responsive frontend user interface.
Node.js	Provides the JavaScript runtime environment for the backend application.
Express.js	Used to build RESTful APIs and handle server-side requests and routing.
MySQL	Used as the relational database for storing users, properties, agents, enquiries, leads, and other application data.
Map box	Provides interactive maps, property locations, markers, and location-based search functionality.
360° Viewer	Enables users to explore properties through immersive 360-degree virtual tours.
HTML5	Provides the basic structure of web pages and application components.
CSS3	Used for styling, responsive layouts, animations, and visual presentation.
JavaScript	Provides application logic and interactive functionality across the platform.
Git	Used for source-code version control and tracking project changes.
GitHub	Used for remote code hosting, collaboration, backup, and project management.
Visual Studio Code	Used as the primary integrated development environment for developing and managing the project.
Postman	Used to test and debug backend REST APIs during development.
Npm	Used for managing Node.js packages and project dependencies.

Development Platform

The application is designed as a **web-based platform** accessible through modern browsers on desktop and mobile devices. The frontend communicates with the backend through RESTful APIs, while the backend interacts with the MySQL database and external services such as map box.

Guide Name and Signature (remarks if any) _____